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Course Name – Data Analytics With GenAI

Subject – PowerBI

Topic – Publishing & Sharing Reports in Power BI

Q1. Explain why publishing and sharing reports is considered more important than just building dashboards in Power BI Desktop.

Building a dashboard in Power BI Desktop is only the first step. A dashboard that stays on one person's computer has no real business value because no one else can see or use it. Publishing and sharing is what turns a local file into an actual business tool. When you publish a report, it becomes accessible to decision-makers, managers, and teams across the organization, anytime, from any device. It enables collaboration, keeps everyone on the same page with live data, and allows organizations to act on insights quickly. Without sharing, even the most beautifully built report is useless to the business.

Q2. Define Power BI Service and explain its role in comparison to Power BI Desktop in the reporting lifecycle.

Power BI Desktop is a free Windows application used for building reports ,connecting to data sources, transforming data, creating visuals, and designing the overall layout. It's essentially the development environment.

Power BI Service (app.powerbi.com) is the cloud-based platform where those reports are published, shared, and consumed. It handles the distribution side ,you can schedule data refreshes, create dashboards by pinning visuals from reports, manage workspaces, set up access control, and share content with colleagues or the entire organization.

In short, Desktop is where you *build*, and Service is where you *deploy and share*. One without the other is incomplete in a real-world workflow.

Q3. What does publishing a report mean in Power BI? List any two outcomes that occur after a report is published.

Publishing a report means uploading your .pbix file from Power BI Desktop to Power BI Service. This process transfers both the report (visuals and pages) and the underlying dataset to your selected workspace in the cloud.

Two outcomes after publishing:

1. The report becomes available in your Power BI Service workspace, where authorized users can view and interact with it through a browser without needing Power BI Desktop installed.

2. A separate dataset is automatically created in the workspace, which can be used independently to build new reports or configure scheduled data refreshes.

Q4. List and explain the steps involved in publishing a report from Power BI Desktop to Power BI Service in the correct sequence.

First, you build and finalize your report in Power BI Desktop. Once it's ready, you go to the **Home** tab in the ribbon and click the **Publish** button. Power BI will prompt you to sign in with your Microsoft/organizational account if you're not already logged in.

After signing in, a dialog box appears asking you to select the destination workspace — this could be "My Workspace" or any shared workspace you have access to. You select the appropriate workspace and click **Select**.

Power BI then begins uploading the file. Once complete, it shows a success message with a link to open the report directly in the browser. Clicking that link takes you straight to your published report in Power BI Service.

Q5. Differentiate between Report and Dashboard in Power BI based on structure and usage.

A **Report** in Power BI is a multi-page document built in Power BI Desktop. Each page can contain multiple visuals — charts, tables, slicers, maps, etc. Reports are connected to a single dataset and allow deep interaction like drilling down, filtering, and cross-highlighting. They are meant for detailed analysis.

A **Dashboard** exists only in Power BI Service and is a single-page canvas made by pinning individual visuals from one or more reports. Dashboards give a high-level summary view and support real-time tiles. They cannot be built in Desktop and don't support the same level of interactivity as reports.

So the key distinction is: reports are for exploration and detail, dashboards are for monitoring and quick snapshots.

Q6. Explain the difference between the following three sharing methods in Power BI Service:

- **Direct Report Sharing**
- **Workspace Access**
- **Power BI Apps**

Also mention one use case for each.

Direct Report Sharing — You share a specific report directly with individual users or groups via their email addresses. They receive a link and can view the report. It's simple and quick. *Use case:* Sharing a one-off sales performance report with a few senior managers for a quarterly review.

Workspace Access — You add users to a workspace and assign them a role (Admin, Member, Contributor, or Viewer). They can then access all content within that workspace based on their role. *Use case:* A data team working together on multiple reports for a single business unit — all team members are given Contributor access to the shared workspace.

Power BI Apps — You package an entire workspace's content (reports, dashboards) into an App and publish it to specific users or the whole organization. The audience sees a clean, curated interface and cannot access the underlying workspace. *Use case:* An HR department wants to distribute a monthly workforce analytics report to 500 employees across the company in a controlled, read-only format.

Q7. Describe the four workspace roles available in Power BI and explain how access control is managed using these roles.

Admin — Has full control over the workspace. Can add or remove members, publish apps, edit and delete all content, and manage all settings. Typically the workspace owner or IT admin.

Member — Can add other members (except Admins), publish and update apps, and create, edit, or delete content in the workspace. Suitable for senior team members managing content.

Contributor — Can create, edit, and delete content in the workspace but cannot publish apps or manage membership. Ideal for analysts and report developers who are actively building content.

Viewer — Read-only access. Can view and interact with reports and dashboards but cannot edit anything. Suited for business users who only need to consume reports.

Access control is managed by assigning the appropriate role to each user or security group based on their responsibility. This ensures the principle of least privilege — people only get as much access as they actually need.

Q8. Why are Power BI Apps considered more secure and scalable than direct sharing? Explain this from a governance and enterprise usage perspective.

When you use direct sharing, you share individual reports one at a time with specific users. This becomes hard to manage as the organization grows — you lose track of who has access to what, and if you update a report, you may need to re-share it. There's no clear separation between the development environment and what end-users see.

Power BI Apps solve this from a governance standpoint. With Apps, the workspace is the back-end (where developers work), and the App is the front-end (what business users see). You can update reports in the workspace and re-publish the App without changing who has access. The audience never touches the workspace directly, which protects the source data and report structure.

From an enterprise perspective, Apps support audience-based access, meaning you can define which groups see which content within the same App. This makes it easy to scale to hundreds or thousands of users while maintaining clear control. IT and governance teams have a single point of management instead of chasing individual share links. This also helps with compliance and audit trails, which are critical in regulated industries.

Q9. Explain Row-Level Security (RLS) in Power BI. How does RLS ensure that different users see different data using the same report?

Row-Level Security is a feature in Power BI that restricts which rows of data a particular user can see when they view a report. Even though everyone is looking at the same report, RLS filters the underlying data based on who is logged in.

It works by defining **roles** in Power BI Desktop using DAX filter expressions. For example, you can create a role called "Region_East" with a filter like `[Region] = "East"`. You then map this role to specific users or security groups in Power BI Service.

When a user assigned to the "Region_East" role opens the report, the data model automatically applies that filter in the background. So a sales manager in the East region will only see East region data, while a manager in the West will only see West data — both using the exact same report.

Dynamic RLS takes this further by using a DAX function like `USERPRINCIPALNAME()` which captures the logged-in user's email and matches it against a mapping table in the data, allowing very granular, per-user data filtering without creating separate reports for each person.

Q10. Assume an organization shares reports without using Apps, workspace roles, or RLS. Discuss any three risks or issues that may arise related to security, governance, or data misuse, and explain how Power BI best practices help prevent them.

Risk 1 — Unauthorized Data Access: Without RLS, anyone who can view a report sees all the data in it. A junior employee might view salary data meant only for HR, or a regional manager might see figures from other regions. This is a serious confidentiality issue. Power BI's RLS prevents this by filtering data at the row level based on the user's identity, ensuring each person only sees what they are authorized to see.

Risk 2 — Uncontrolled Sharing and Data Leakage: When reports are shared directly without workspace roles, the recipients can sometimes re-share those reports further. This creates a chain of access that is impossible to track or revoke. Sensitive business data can quickly end up with people who should never have seen it. Using workspace roles and Apps gives admins a centralized way to control and revoke access instantly, and Apps can be configured to prevent further sharing.

Risk 3 — No Audit Trail or Governance: Without structured access management, there is no easy way to know who accessed what data and when. This is a compliance nightmare, especially for organizations in finance, healthcare, or legal sectors where data governance is regulated. Power BI Service, when used with workspace roles and Apps, integrates with Microsoft 365 audit logs and the Power BI Admin portal, which lets governance teams monitor usage, track access, and respond to data incidents with proper evidence.